Geon Choi

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Research Interests

Multimodal AI
Vision-Language Models (VLMs), Vision-Language-Action (VLA) Models

AI for Healthcare
VLMs and Agents for Medical Imaging, Data Synthesis

Computer Vision
Generative Models

Education

Korea Advanced Institute of Science and Technology (KAIST) Seongnam, South Korea
Ph.D. in Artificial Intelligence (Advisor: Prof. Edward Choi) *Feb. 2026 – Feb.2029 (Expected)*

Korea Advanced Institute of Science and Technology (KAIST) Seongnam, South Korea
M.S. in Artificial Intelligence (Advisor: Prof. Edward Choi) *Feb. 2024 – Feb. 2026*

Pohang University of Science and Technology (POSTECH) Pohang, South Korea
B.S. in Electrical Engineering, GPA: 4.04/4.3 (Summa Cum Laude, Ranked 2nd in Department) *Feb. 2020 – Feb. 2024*

Publications

C: Conference **J**: Journal **P**: Preprint *****: Equal Contribution

[C.3] Lunguage: A Benchmark for Structured and Sequential Chest X-ray Interpretation
Jong Hak Moon, Geon Choi, Paloma Rabaey, Min Gwan Kim, Hyuk Gi Hong, Jung-Oh Lee, Hangyul Yoon, Eun Woo Doe, Jiyoun Kim, Harshita Sharma, Daniel C. Castro, Javier Alvarez-Valle, Edward Choi
Conference on Health, Inference, and Learning (CHIL), 2026

- Built an LLM-based framework that structures and evaluates VLM-generated radiology reports.
- [ArXiv] [Code] [Dataset]

[C.2] Instruction-Guided Lesion Segmentation for Chest X-rays with Automatically Generated Large-Scale Dataset
Geon Choi*, Hangyul Yoon*, Hyunju Shin, Hyunki Park, Sang Hoon Seo, Eunho Yang, Edward Choi
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

- Built MIMIC-ILS, a 1M-scale automatically generated dataset for lesion segmentation, by orchestrating LLMs with vision models (diffusion-based editing, segmentation, detection).
- Developed ROSALIA, an instruction-guided lesion segmentation model that integrates a VLM with SAM, trained on MIMIC-ILS.
- [ArXiv] [Code] [Dataset] [Model]

[C.1] CXReasonBench: A Benchmark for Evaluating Structured Diagnostic Reasoning in Chest X-rays
Hyungyung Lee, Geon Choi, Jung-Oh Lee, Hangyul Yoon, Hyuk Gi Hong, Edward Choi
Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2025

- **Spotlight Presentation (63/1995=3.2%)**
- Designed a multi-stage benchmark for evaluating the medical reasoning capabilities of VLMs.
- Built the information-extraction pipeline for X-ray image quality assessment sub-tasks (inclusion, rotation).
- [ArXiv] [Code] [Dataset]

[P.1] CheXpercept: A Benchmark for Evaluating Expert-Level Lesion Perception in Chest X-rays
Geon Choi, Hangyul Yoon, Nalee Kim, Jeong Yun Jang, Hyunju Shin, Hyunki Park, Sang Hoon Seo, Edward Choi
Under Review

- Designed a benchmark for evaluating VLMs' lesion multi-level perception capabilities in chest X-rays.
- Built the benchmark semi-automatically by leveraging models such as SAM, reducing manual annotation by physicians.
- [ArXiv]

Experience

KAIST AI Research Internship (KAIRI) — Jun. 2023 – Aug. 2023
BioImaging, Signal Processing and Learning Lab, KAIST (Advisor: Prof. Jongchul Ye) — *Daejeon, South Korea*

- Research Topic: Generative Models for Solving Inverse Problems in Medical Imaging.

Undergraduate Research Internship — Jan. 2023 – Jun. 2023
Bio Optics and Acoustics Lab, POSTECH (Advisor: Prof. Chulhong Kim) — *Pohang, South Korea*

- Research Topic: Computer-Aided Diagnosis System for Breast Ultrasonography.

Honors and Awards

KAIST Scholarship — 2024 – Present
Tuition Scholarship — *KAIST*

KEC Scholarship — 2023
Merit Scholarship for Outstanding EE Student ($2k) — *KEC Foundation*

National Science & Technology Scholarship — 2022 – 2023
Full Tuition Scholarship — *KOSAF*

POSTECH Jigok Scholarship — 2020 – 2021
Full Tuition Scholarship — *POSTECH*

POSTECH Presidential Fellowship — 2020 – 2023
Merit Fellowship for Exceptional Freshman ($10k) — *POSTECH*

Teaching

Course Assistant, Digital System Design (EECE273) — 2022
Student Mentor — *POSTECH*

Services

KoSAIM Summer School Hands-on Session — 2024 – 2025
Instructor — *KoSAIM*

- Taught chest X-ray classification with CNN for medical AI researchers and medical students in South Korea.

Languages

English: Professional working proficiency
Korean: Native

Technical Skills

Programming Languages: Python, C, MATLAB
Deep Learning: PyTorch, HuggingFace Transformers
Medical Imaging: pydicom, SimpleITK, nibabel, OpenCV, scikit-image
Tools & Infrastructure: Git, Linux, TensorBoard